

## Fungal Diversity and Plant Growth Promotion of Endophytic Fungi from Six Halophytes in Suncheon Bay

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Endophytic fungi were isolated from roots of six halophytes in Suncheon Bay. The endophytic fungi of 35 species isolated from halophytes were identified by internal transcribed spacer (ITS) containing the ITS1, 5.8s, and ITS2 regions. All fungal strains were analyzed to diversity at the genus level. Fungal culture filtrates (FCF) of endophytic fungi were treated to Waito-c rice (WR) seedling for plant growth-promoting verification. It was confirmed that fungal strain Sj-2-2 provided plant growth promotion (PGP) to WR seedling. Then, PGP of *Suaeda japonica* was confirmed by treating culture filtrate of Sj-2-2. As a result, it was verified that culture filtrate of Sj-2-2 had more advanced PGP than positive control when treated to *S. japonica*. The secondary metabolites involved in culture filtrate of Sj-2-2 were identified by HPLC and GC-MS SIM analysis. The presence of physiologically bioactive gibberellins (GAs) and other inactive GAs in culture filtrate of Sj-2-2 was detected. The molecular analysis of sequences of Sj-2-2 showed the similarity to *Penicillium* sp. of 99% homology. The PGP of Sj-2-2 as well as symbiosis between endophytic fungi and halophytes growing naturally in salt marsh was confirmed. Sj-2-2 was identified as a new fungal strain producing GAs by molecular analysis of sequences. Consequently, the Sj-2-2 fungal strain was named as *Penicillium* sp. Sj-2-2. In this study, the diversity of endophytic fungi isolated from roots of halophytes in salt marsh and the PGP of a new gibberellin-producing fungal strain were confirmed.

**Keywords:** Endophytic fungi, fungal diversity, plant growth promotion, gibberellin, halophyte, salt marsh

Marshes as ecotones between terrestrial and water ecosystems are inhabited by various biospecies [23] and function in environmental cleanup, as a supply of surface water, in flow control, and in the storage and recharge of underground water. Salt marshes harbor a high diversity of microorganisms as well as various hydrophytes [17]. Examinations of microbial abundance and diversity have been done using cluster analysis of soil isolates [5, 20]. However, no valid experiment of endophytic fungi related with wild halophytes in salt marshes has been reported.

Endophytic fungi live in symbiosis with plants and play an important role in plant growth. Fungi produce a wide variety of plant hormones such as gibberellins (GAs), abscisic acid (ABA), and auxin (IAA). Research on biological control using these hormones has increased [4, 12, 16]. The phytohormone GA, a diterpenoid complex, controls the growth of plants, and promotes flowering, stem elongation, seed germination, and ripening [8, 25]. *Gibberella fujikuroi*, which causes bakanae, was discovered in the 1920s and its active component was isolated in 1935 [19]. GA is produced by *G. fujikuroi*, *Phaeosphaeria* sp. L487, *Neurospora crassa*, *Pseudomonas* sp., *Paceloma manihoticola*, *Aspergillus* sp., and *Penicillium* sp., *Penicillium minioluteum*, *Paecilomyces formosus*, among others; it is produced by bacteria, plants, and fungi [1, 8–11, 13, 14, 16, 24]. One hundred thirty-six kinds of GAs have been identified as secondary metabolites; GA<sub>1</sub>, GA<sub>3</sub>, GA<sub>4</sub>, and GA<sub>7</sub> have PGP activity [2, 3, 11, 15, 16]. There have been various studies of plant growth related to GAs [6, 25].

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# Supplementary data for this paper are available on-line only at <http://jmb.or.kr>.

In this study, endophytic fungi related to and symbiotic with wild halophytes in Suncheon Bay were screened morphologically as different endophytic fungi. Identification was conducted with ITS sequences to identify endophytic fungi, and the diversity of the endophytic fungi was examined. A bioassay was also conducted using the WR seedling and halophytes to clarify the effects on PGP with endophytic fungi. The secondary metabolites from isolated endophytic fungi were analyzed to clarify the features of endophytic fungi.

## MATERIALS AND METHODS

### Plant Materials

Healthy roots of halophytes were collected from a salt marsh of Suncheon Bay in South Korea. The local sites and scientific names of six halophytes are listed in Table 1.

### Preparation and Growth Conditions of the Sample

The root samples of halophytes were treated with Tween 80 solution for 10 min, and the surface was sterilized by bleaching using 1% (w/v) perchloric acid solution. The sterilized roots were cut into 3 cm pieces and incubated at 25°C until emergence of fungi from inside of root pieces occurred during culture on hagem minimal medium containing streptomycin [26, 27]. The fungal strains isolated from the root pieces were subcultured on PDA. Czapek broth medium (CBM) containing 1% (w/v) glucose and peptone were used for production of secondary metabolites [7]. Wild-type strain *G. fujikuroi* provided by the Korean Culture Center of Microorganisms (KCCM) was used as a positive control in this study.

### DNA Extraction, PCR Amplification, and Identification of Fungal Strains

The subculture of fungal strains was incubated in PDA for 7–10 days. The mycelia of fungal strains were incubated into an Erlenmeyer flask containing 50 ml of potato dextrose broth medium in a shaking incubator for 7 days at 120 rpm at 28°C. The lyophilized fungal strains were obtained by vacuum filtration. The fungal genomic DNA was extracted with a DNeasy plant mini kit (QIAGEN, Valencia, CA, USA). The fungal strains were identified by sequencing the ITS region with universal primers ITS-1 (5'-TCC GTA GGT GAA CCT GCG G-3') and ITS-4 (5'-TCC TCC GCT TAT TGA TAT GC-3'). Fifty microliters of a total PCR mixture was used. The mixture contained 5 µl (v/v) of dNTPs and Ex-Taq buffer, 4 µl (v/v) of each primer, 1 µl (v/v) of DNA sample, 0.25 µl (v/v) of Ex-Taq polymerase, and 30.75 µl (v/v) of autoclaved deionized distilled

water. The reaction cycles consisted of 2 min of initial denaturation (95°C), 35 cycles by 2 min of denaturation (95°C), 1 min of annealing (55°C), 1 min of extension (72°C), and 7 min of final extension (72°C). The 5 µl (v/v) PCR products were observed by agarose gel electrophoresis with ethidium bromide (EtBr) staining. The products were purified and sequenced with a QIAquick PCR purification kit (QIAGEN), ABI PRISM BigDye Terminator Cycle Sequencing Kit (PE Biosystems, Foster, CA, USA), and ABI 310 DNA sequencer (Perkin Elmer).

### Phylogenetic Analysis of Fungal Strains

The BLAST program was searched to look for nucleotide sequences of rDNA-ITS regions. The obtained sequences were aligned by ClustalW with the MEGA version 4.0 software program [21]. ClustalW was used to align the obtained sequences and to generate a neighbor-joining (NJ) tree. A statistical support for the nodes in the phylogenetic tree was obtained with bootstrap replication (1,000 replications).

### Data Analysis with Statistical Treatment

The diversity of endophytic fungi isolated from six halophytes was calculated by the Shannon diversity index ( $H'$ ) following to the formula

$$H' = -\sum_{i=1}^k P_i \ln P_i$$

where  $k$  is the total number of fungal species encountered and  $P_i$  is the proportion of all individuals in a sample that species  $i$  contributes to the total [22]. The data of all graphs were analyzed statistically for standard deviation using Microsoft Excel. Statistical analysis of PGP was conducted using one-way analysis of variance (ANOVA). A Duncan's multiple range test (DMRT) was conducted to analyze significant differences ( $P < 0.05$ ) among the different groups (using SPSS 18.0 for Windows; SPSS, Chicago, IL, USA).

### Screening of FCFs for Plant Growth Promotion on WR Seedling

The fungal strains were incubated for 7 days at 25°C and 180 rpm in CBM. The harvested mycelia and FCFs were stored at -70°C and lyophilized. The lyophilized FCFs were mixed with 1 ml of autoclaved distilled water. The seeds of sterilized WR were treated overnight with 20 ppm uniconazol (Sumitomo Heavy Chemical Co., Takarazuka, Japan) to minimize the activity of seed coat GA. The seeds of the treated WR were washed and soaked in sterile distilled water until sprouts emerged. WR seedlings were transplanted in glass tubes containing 0.6% water-agar and grown in a growth chamber. Ten microliters of a concentrated solution of FCF was applied to the apical meristem of WR seedlings at the two-leaf stage. Shoot length (SL) and plant length (PL) of WR seedlings were observed a week after the seedlings were treated with distilled water, CBM, and

**Table 1.** Details of sampling sites and halophytes species collected in this study.

| Scientific names                                   | Plant code | Site of collection           | Habitat    |
|----------------------------------------------------|------------|------------------------------|------------|
| <i>Artemisia fukudo</i> Makino                     | Af         | E 127° 30' 51"/N 34° 52' 47" | Halophytic |
| <i>Carex scabrifolia</i> Steud.                    | Cs         | E 127° 30' 52"/N 34° 52' 45" | Halophytic |
| <i>Kochia scoparia</i> var. <i>littorea</i> Makino | Ks         | E 127° 30' 50"/N 34° 52' 48" | Halophytic |
| <i>Phragmites communis</i> Trin.                   | Pc         | E 127° 30' 50"/N 34° 52' 47" | Halophytic |
| <i>Suaeda australis</i> (R.Br.) Moq.               | Sa         | E 127° 30' 53"/N 34° 52' 42" | Halophytic |
| <i>Suaeda japonica</i> Makino                      | Sj         | E 127° 30' 52"/N 34° 52' 44" | Halophytic |

culture filtrate of *G. fujikuroi*. Negative and positive controls were treated as controls to deduce the SL and PL of seedlings.

#### Bioassay on *S. japonica* Seedling

The seeds of *S. japonica* were collected from Suncheon Bay in South Korea. The surface-sterilized *S. japonica* seeds were soaked in distilled water for germination and then treated with uniconazol. *S. japonica* seedlings were cultivated in a growth chamber at 25°C. The two-leaves stage of *S. japonica* seedlings were treated with lyophilized FCF (10 µl) of fungal strain Sj-2-2. The SL and PL were measured after 14 days and compared with control treatments.

#### Extraction and Quantification of GAs from FCFs

GA was extracted from FCF after incubation for 7 days according to an established procedure [18]. The extracted GAs were analyzed with reverse-phase C<sub>18</sub> HPLC. The fractions were collected and prepared for GC-MS with selected ion monitoring (SIM) (6890N

network GC system, and 5973 network mass selective detector; Agilent Technologies, Palo Alto, CA, USA). The GC-MS data were collected and analyzed. The three major ions of the supplemented [<sup>2</sup>H<sub>2</sub>] GA internal standards (obtained from Prof. Lewis N. Mander, Australian National University, Canberra, Australia) and GAs were monitored simultaneously. Hydrocarbon standards to calculate the Kovats Retention Index (KRI) value were used to determine the retention time, and peak area ratios of non-deuterated (extracted) GAs to deuterated GAs were used to quantify the GAs [3, 8, 15, 16].

## RESULTS

### Identification and Phylogenetic Analysis of Endophytic Fungi

The nucleotide sequences of endophytic fungi were registered in the GenBank database of the National Center for

**Table 2.** Identification of endophytic fungi isolated in this study.

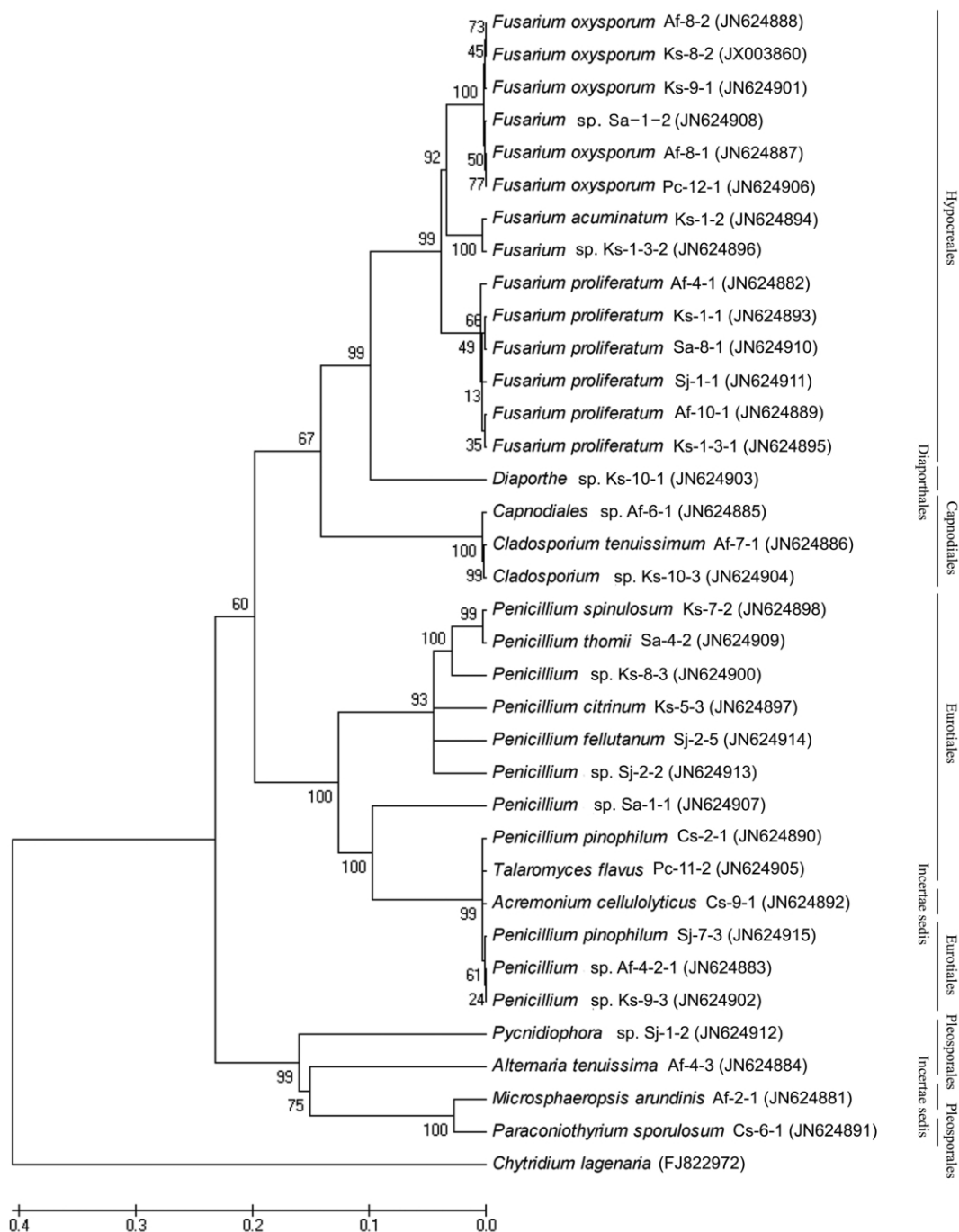
| Plant code | Fungal isolates | Closest relative based on sequence homology   | Similarity (%) | Accession No. |
|------------|-----------------|-----------------------------------------------|----------------|---------------|
| Af         | Af-2-1          | <i>Microsphaeropsis arundinis</i> (EF094556)  | 99             | JN624881      |
|            | Af-4-1          | <i>Fusarium proliferatum</i> (GQ924905)       | 99             | JN624882      |
|            | Af-4-2-1        | <i>Penicillium</i> sp. (HQ671193)             | 99             | JN624883      |
|            | Af-4-3          | <i>Alternaria tenuissima</i> (FJ949080)       | 100            | JN624884      |
|            | Af-6-1          | <i>Capnodiales</i> sp. (HQ207047)             | 99             | JN624885      |
|            | Af-7-1          | <i>Cladosporium tenuissimum</i> (HM776419)    | 99             | JN624886      |
|            | Af-8-1          | <i>Fusarium oxysporum</i> (HQ451894)          | 99             | JN624887      |
|            | Af-8-2          | <i>Fusarium oxysporum</i> (HQ451890)          | 99             | JN624888      |
|            | Af-10-1         | <i>Fusarium proliferatum</i> (HQ332533)       | 100            | JN624889      |
|            | Af-10-2         | <i>Fusarium proliferatum</i> (HQ332533)       | 100            | JN624889      |
| Cs         | Cs-2-1          | <i>Penicillium pinophilum</i> (HM469418)      | 100            | JN624890      |
|            | Cs-6-1          | <i>Paraconiothyrium sporulosum</i> (GU566257) | 100            | JN624891      |
|            | Cs-9-1          | <i>Acremonium cellulolyticus</i> (GU479898)   | 99             | JN624892      |
|            | Cs-9-2          | <i>Acremonium cellulolyticus</i> (GU479898)   | 99             | JN624892      |
| Ks         | Ks-1-1          | <i>Fusarium proliferatum</i> (FJ648201)       | 99             | JN624893      |
|            | Ks-1-2          | <i>Fusarium acuminatum</i> (AB587001)         | 99             | JN624894      |
|            | Ks-1-3-1        | <i>Fusarium proliferatum</i> (GQ924896)       | 100            | JN624895      |
|            | Ks-1-3-2        | <i>Fusarium</i> sp. (HQ631036)                | 99             | JN624896      |
|            | Ks-5-3          | <i>Penicillium citrinum</i> (HQ671192)        | 99             | JN624897      |
|            | Ks-7-2          | <i>Penicillium spinulosum</i> (HM469405)      | 99             | JN624898      |
|            | Ks-8-2          | <i>Fusarium oxysporum</i> (GU566240)          | 92             | JX003860      |
|            | Ks-8-3          | <i>Penicillium</i> sp. (HQ631040)             | 99             | JN624900      |
|            | Ks-9-1          | <i>Fusarium oxysporum</i> (HQ451892)          | 99             | JN624901      |
|            | Ks-9-3          | <i>Penicillium</i> sp. (FJ795356)             | 99             | JN624902      |
|            | Ks-10-1         | <i>Diaporthe</i> sp. (DQ145734)               | 99             | JN624903      |
|            | Ks-10-3         | <i>Cladosporium</i> sp. (HQ671188)            | 99             | JN624904      |
|            | Pc-11-2         | <i>Talaromyces flavus</i> (HQ191279)          | 99             | JN624905      |
|            | Pc-12-1         | <i>Fusarium oxysporum</i> (HQ451893)          | 99             | JN624906      |
| Sa         | Sa-1-1          | <i>Penicillium</i> sp. (HM849046)             | 93             | JN624907      |
|            | Sa-1-2          | <i>Fusarium</i> sp. (GQ470679)                | 99             | JN624908      |
|            | Sa-4-2          | <i>Penicillium thomii</i> (AF034448)          | 99             | JN624909      |
|            | Sa-8-1          | <i>Fusarium proliferatum</i> (GQ924898)       | 100            | JN624910      |
| Sj         | Sj-1-1          | <i>Fusarium proliferatum</i> (GQ924897)       | 100            | JN624911      |
|            | Sj-1-2          | <i>Pycnidophora</i> sp. (GU595028)            | 98             | JN624912      |
|            | Sj-2-2          | <i>Penicillium</i> sp. (HQ225717)             | 99             | JN624913      |
|            | Sj-2-5          | <i>Penicillium fellutanum</i> (HM469425)      | 100            | JN624914      |
|            | Sj-7-3          | <i>Penicillium pinophilum</i> (HM469418)      | 99             | JN624915      |

The endophytic fungi were cultured at 180 rpm and 25°C in the dark. Universal primer ITS1 and ITS4 were used for identification of endophytic fungi in PCR reactions. Amplified fragments were analyzed to compare the ITS regions with the BLAST program. The nucleotide sequences of endophytic fungi have been deposited in the NCBI database.

Biotechnology Information (Accession numbers JN624881–JN624898, JX003860, JN624900–JN624915) (Table 2). Thirty-five endophytic fungi isolated from the halophytes were confirmed to belong to Ascomycota. It was confirmed that *Penicillium* sp. and *Fusarium* sp. were the main endophytic fungi; then phylogenetic analysis was carried out (Fig. 1).

### Fungal Diversity of Endophytic Fungi Isolated from Halophytes

Thirty-five culturable fungal strains were isolated from roots of six halophytes based on colony morphologies. All endophytic fungi from halophytes belonged to 11 genera based on molecular identification. The Shannon's diversity



**Fig. 1.** Phylogenetic analysis of endophytic fungi using rDNA-ITS sequences.

A dendrogram was constructed using 36 taxa (35 reference sequences and 1 clone) with the neighbor-joining method (1,000 bootstrap replications). Phylogenetic analysis showed that all isolated fungal strains belong to the phylum Ascomycota. *Chytridium lagenaria* belonging to the Chytridiomycota was used as the outgroup for rooting the phylogenetic tree.



indices of endophytic fungi were greater in *A. fukudo* (1.5811), and lowest in *P. communis* and *S. australis* (0.6932). The endophytic fungi showed different relative frequencies in all halophytes (Table 3).

#### Screening for Plant Growth-Promoting Activity of FCFs on WR Seedling

The culture filtrates of all endophytic fungi were applied to WR seedlings to verify PGP activity. A result of the culture filtrate treatment of the Sj-2-2 fungal strain indicated

13.033 cm of SL and 21.7 cm of PL and confirmed PGP activity. The culture filtrate treatment of the Sj-2-2 fungal and *G. fujikuroi* strains confirmed the similarity of PGP between these two strains (Table 4).

#### Bioassay of Sj-2-2 Strain for Plant Growth-Promotion on *S. japonica*

The SL and PL of *S. japonica* seedlings treated with Sj-2-2 FCF were measured after 14 days of culture filtrate application. The PGP capacity of *S. japonica* seedling by

**Table 3.** Distribution and diversity index of endophytic fungi isolated from six halophytes in Suncheon Bay.

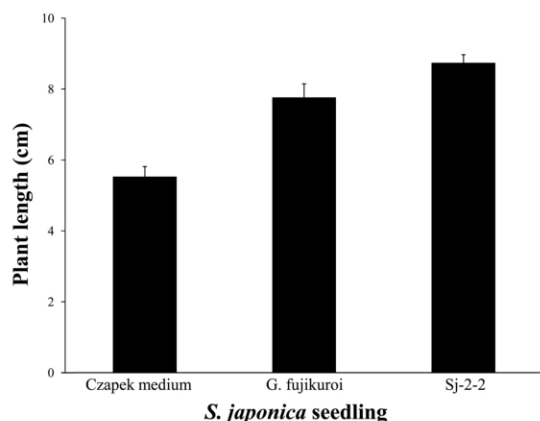
| Fungal taxonomy           | Af     | Cs     | Ks     | Pc     | Sa     | Sj     | Total |
|---------------------------|--------|--------|--------|--------|--------|--------|-------|
| <i>Acremonium</i>         |        | 1      |        |        |        |        | 1     |
| <i>Alternaria</i>         | 1      |        |        |        |        |        | 1     |
| <i>Capnodiales</i>        | 1      |        |        |        |        |        | 1     |
| <i>Cladosporium</i>       | 1      |        | 1      |        |        |        | 2     |
| <i>Diaporthe</i>          |        |        | 1      |        |        |        | 1     |
| <i>Fusarium</i>           | 4      |        | 5      | 1      | 2      | 1      | 13    |
| <i>Microsphaeropsis</i>   | 1      |        |        |        |        |        | 1     |
| <i>Paraconiothyrium</i>   |        | 1      |        |        |        |        | 1     |
| <i>Penicillium</i>        | 1      | 1      | 5      |        | 2      | 3      | 12    |
| <i>Pycnidophora</i>       |        |        |        |        |        | 1      | 1     |
| <i>Talaromyces</i>        |        |        |        | 1      |        |        | 1     |
| Total                     | 9      | 3      | 12     | 2      | 4      | 5      | 35    |
| Shannon's diversity index | 1.5811 | 1.0986 | 1.1437 | 0.6932 | 0.6932 | 0.9503 |       |

**Table 4.** Screening for PGP of WR seedling with FCFs of endophytic fungi isolated from halophytes.

| Fungal isolates        | SL                             | PL                               | Fungal isolates | SL                             | PL                              |
|------------------------|--------------------------------|----------------------------------|-----------------|--------------------------------|---------------------------------|
| D. water*              | 6.07 <sup>klm</sup> ± 0.32     | 13.57 <sup>no</sup> ± 0.85       | Ks-5-3          | 8.83 <sup>bcd</sup> ± 0.45     | 17.43 <sup>bcd</sup> ± 0.78     |
| Medium                 | 7.10 <sup>efghijk</sup> ± 0.66 | 14.70 <sup>ijklmn</sup> ± 0.40   | Ks-7-2          | 9.17 <sup>b</sup> ± 0.45       | 17.17 <sup>bcd</sup> ± 0.25     |
| <i>G. fujikuroi</i> ** | 12.83 <sup>a</sup> ± 0.42      | 21.57 <sup>a</sup> ± 0.71        | Ks-8-2          | 6.50 <sup>ijklm</sup> ± 0.79   | 16.27 <sup>cdefghi</sup> ± 0.51 |
| Af-2-1                 | 7.93 <sup>cdefg</sup> ± 0.67   | 15.47 <sup>efghijk</sup> ± 0.83  | Ks-8-3          | 8.13 <sup>bcd</sup> ± 0.25     | 15.70 <sup>efghijk</sup> ± 1.10 |
| Af-4-1                 | 7.67 <sup>defghi</sup> ± 0.73  | 16.20 <sup>cdefghij</sup> ± 0.46 | Ks-9-1          | 6.00 <sup>klm</sup> ± 0.62     | 14.77 <sup>ijklmn</sup> ± 1.01  |
| Af-4-2-1               | 8.07 <sup>bcdef</sup> ± 0.51   | 15.67 <sup>efghijk</sup> ± 0.72  | Ks-9-3          | 7.90 <sup>cdefgh</sup> ± 0.30  | 15.30 <sup>ghijklm</sup> ± 0.87 |
| Af-4-3                 | 7.87 <sup>cdefgh</sup> ± 0.42  | 16.10 <sup>dceghij</sup> ± 0.66  | Ks-10-1         | 7.53 <sup>efghij</sup> ± 0.71  | 16.17 <sup>defghij</sup> ± 0.81 |
| Af-6-1                 | 7.30 <sup>efghij</sup> ± 0.75  | 14.67 <sup>ijklmn</sup> ± 0.81   | Ks-10-3         | 9.00 <sup>bc</sup> ± 0.66      | 17.67 <sup>bc</sup> ± 0.40      |
| Af-7-1                 | 7.30 <sup>efghij</sup> ± 0.20  | 14.10 <sup>lmn</sup> ± 0.46      | Pc-11-2         | 9.07 <sup>bc</sup> ± 0.51      | 17.07 <sup>bcd</sup> ± 0.42     |
| Af-8-1                 | 6.57 <sup>ijklm</sup> ± 0.64   | 14.70 <sup>ijklmn</sup> ± 0.62   | Pc-12-1         | 6.60 <sup>ijklm</sup> ± 0.70   | 15.13 <sup>hijklm</sup> ± 0.40  |
| Af-8-2                 | 7.43 <sup>efghij</sup> ± 0.49  | 15.17 <sup>hijklm</sup> ± 0.59   | Sa-1-1          | 8.27 <sup>bcd</sup> ± 0.51     | 15.83 <sup>efghijk</sup> ± 1.82 |
| Af-10-1                | 5.87 <sup>lm</sup> ± 0.49      | 12.40 <sup>0</sup> ± 0.70        | Sa-1-2          | 8.43 <sup>bcd</sup> ± 0.55     | 16.67 <sup>bcd</sup> ± 1.05     |
| Cs-2-1                 | 6.33 <sup>ijklm</sup> ± 0.71   | 15.17 <sup>hijklm</sup> ± 0.56   | Sa-4-2          | 5.47 <sup>m</sup> ± 0.60       | 13.87 <sup>mn</sup> ± 0.35      |
| Cs-6-1                 | 6.53 <sup>ijklm</sup> ± 1.06   | 14.37 <sup>klmn</sup> ± 1.25     | Sa-8-1          | 6.83 <sup>ghijkl</sup> ± 0.93  | 15.00 <sup>ijklmn</sup> ± 0.44  |
| Cs-9-1                 | 6.80 <sup>ghijkl</sup> ± 0.36  | 15.73 <sup>efghijk</sup> ± 0.61  | Sj-1-1          | 7.13 <sup>efghijk</sup> ± 0.45 | 15.63 <sup>efghijk</sup> ± 0.45 |
| Ks-1-1                 | 8.20 <sup>bcd</sup> ± 0.50     | 15.77 <sup>efghijk</sup> ± 0.60  | Sj-1-2          | 6.70 <sup>hijkl</sup> ± 0.36   | 15.40 <sup>ghijklm</sup> ± 0.78 |
| Ks-1-2                 | 5.63 <sup>lm</sup> ± 0.95      | 14.43 <sup>klmn</sup> ± 1.25     | Sj-2-2          | 13.03 <sup>a</sup> ± 0.42      | 21.70 <sup>a</sup> ± 0.82       |
| Ks-1-3-1               | 7.47 <sup>efghij</sup> ± 0.73  | 15.53 <sup>ghijkl</sup> ± 0.78   | Sj-2-5          | 8.50 <sup>bcd</sup> ± 0.53     | 17.80 <sup>b</sup> ± 0.36       |
| Ks-1-3-2               | 7.43 <sup>efghij</sup> ± 1.16  | 15.40 <sup>ghijklm</sup> ± 0.30  | Sj-7-3          | 7.37 <sup>efghij</sup> ± 0.55  | 16.83 <sup>bcd</sup> ± 1.24     |

Ten microliters of lyophilized FCFs was applied to WR seedlings. The SL and PL of the seedlings were measured after 7 days of treatment. According to DMRT ( $p < 0.05$ ), the different letters in a row indicate significant differences. The letters indicate that values are not significantly different. Data are expressed as mean value ± standard deviation.

\*Negative control (Distilled water), \*\*Positive control (Wild-type *G. fujikuroi*).



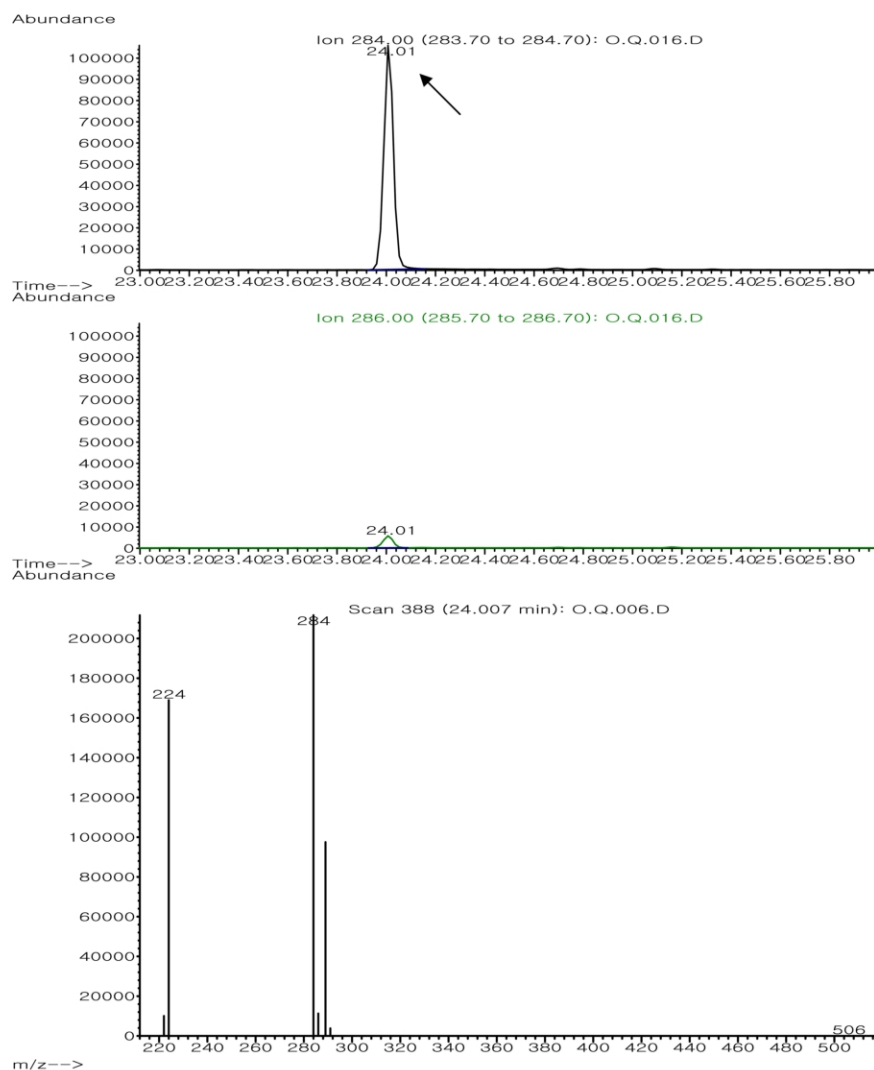
**Fig. 2.** Bioassay of FCFs on *S. japonica* seedlings of endophytic fungi.

Ten microliters of lyophilized FCFs was treated on *S. japonica* seedlings. The SL and PL of *S. japonica* seedlings were measured after 14 days of treatment.

the Sj-2-2 fungal strain was compared with that of concentrated culture filtrate of CBM and that of wild-type *G. fujikuroi*, respectively. As CBM contains nutrients, its control was used to observe if the nutrients affects shoot elongation. The seedlings treated with the SL of Sj-2-2 culture filtrate were 8.74 cm, which were higher than those with *G. fujikuroi* culture filtrate (7.76 cm) and those with Czapek broth (5.53 cm) (Fig. 2).

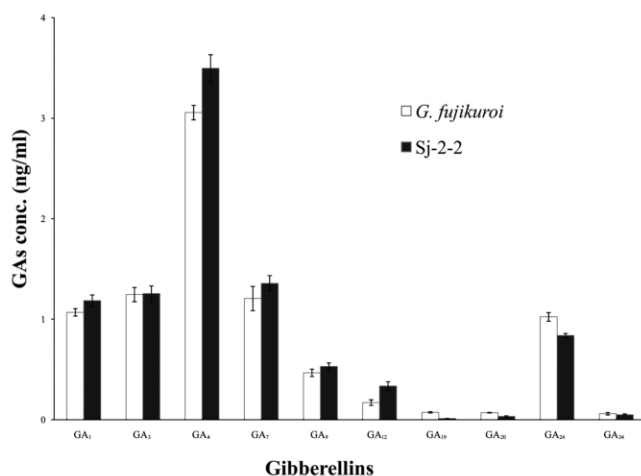
### Quantitative Analysis of Gibberellins from Culture Filtrates of Sj-2-2 Fungal Strain

GA, the plant hormone produced by endophytic fungi isolated from halophytes, was analyzed with HPLC and GC-MS. Consequently, a variety of GAs were confirmed from the culture filtrate of the Sj-2-2 fungal strain; the result of the GC-MS SIM analysis showed that Sj-2-2 produced GA<sub>1</sub> (1.185 ng/ml), GA<sub>3</sub> (1.255 ng/ml), GA<sub>4</sub> (3.497 ng/ml)



**Fig. 3.** GC-MS SIM spectra for GA<sub>4</sub> in FCF of *Penicillium* sp. Sj-2-2.

The arrow refers to the peak of fungal GA<sub>4</sub>, which corresponds with that of internal standard GA<sub>4</sub>.



**Fig. 4.** Concentrations of GAs involved in the FCFs of the Sj-2-2 strain and *G. fujikuroi*.

GC-MS SIM analysis of culture filtrate extracts from the Sj-2-2 fungal strain detected four bioactive GAs. Sj-2-2 showed the presence of bioactivity of GA<sub>1</sub>, GA<sub>3</sub>, GA<sub>4</sub>, GA<sub>7</sub>, and other inactive GAs.

(Fig. 3), GA<sub>7</sub> (1.357 ng/ml), other inactive GA<sub>9</sub> (0.530 ng/ml), GA<sub>12</sub> (0.335 ng/ml), GA<sub>19</sub> (0.011 ng/ml), GA<sub>20</sub> (0.033 ng/ml), GA<sub>24</sub> (0.838 ng/ml), and GA<sub>34</sub> (0.049 ng/ml) (Fig. 4). It was confirmed that Sj-2-2 produced as much GA<sub>1</sub>, GA<sub>3</sub>, and GA<sub>9</sub> as *G. fujikuroi*, and produced much more GA<sub>4</sub>, GA<sub>7</sub>, and GA<sub>12</sub>.

## DISCUSSION

Thirty-five endophytic fungi were isolated from roots of six halophytes and identified by phylogenetic analysis and ITS sequences containing the ITS1, 5.8s, and ITS2 regions. All endophytic fungi were identified at the genus level, containing *Acremonium*, *Alternaria*, *Capnodiales*, *Cladosporium*, *Diaporthe*, *Fusarium* (*Gibberella*), *Microsphaeropsis*, *Paraconiothyrium*, *Penicillium*, *Pycnidophora*, and *Talaromyces*. Of all the endophytic fungi isolated, the genus *Fusarium* and *Penicillium* were the most abundantly distributed in six halophytes. The phylum of various endophytic fungi was confirmed to be included in Ascomycota. This report showed the existence of endophytic fungi at the roots of various halophytes growing in salt marsh.

To confirm the presence of secondary metabolites, a screening of microbial culture filtrates was used to identify biologically active molecules. In this study, PGP hormones were detected in the culture filtrates of endophytic fungi isolated from the roots of halophytes using WR seedlings. The use of WR, a dwarf rice mutant with reduced GA biosynthesis, was highly efficient [8, 14, 15]. Presently, seeds of WR were treated with uniconazol as a GA biosynthesis retardant. The Sj-2-2 fungal strain, which has

strain PGP activity, was identified using WR seedlings. Then, the PGP activity of halophyte was confirmed by *S. japonica* seedling. The Sj-2-2 fungal strain proved the advanced PGP activity of WR and *S. japonica* seedling.

DNA sequence analysis provides various and important information for recent experiments. Many rDNA genes are highly conserved in the taxonomic side for identification. ITS genes are usually used to conduct identification of fungal strains. Based on these facts, the identification of fungal strains was conducted with 5.8S gene, ITS1, and ITS2 regions [8, 15, 16]. Phylogenetic classification analysis with BLAST database searching has become very important for the identification and tracking of genetic evolution. This method is based on statistical analyses like bootstrap tests, and maximum DNA sequence homology provides the most valuable information. Bootstrap consensus is able to construct a phylogenetic tree with maximum sequence replication. Bootstrapping can be applied to a NJ tree to identify various endophytic fungi [8, 15]. The homology and relationship were confirmed through phylogenetic analysis with various endophytic fungi. The Sj-2-2 fungal strain with phylogenetic analysis was named *Penicillium* sp. Sj-2-2 (JN624913).

The GC-MS SIM has the ability to analyze highly complex mixtures and to detect compounds of different classes simultaneously [3, 8, 12, 17], and so was used for culture filtrate analysis of the Sj-2-2 fungal strain. GC-MS SIM is useful to analyze a number of compounds and is often used in plant research [8, 16]. Because of its reliability, GC-MS SIM was used in quantitative analysis of various phytohormones.

In summary, various endophytic fungi were isolated from the roots of six halophytes growing spontaneously in a salt marsh and their relationships were confirmed. A study of PGP activity of *Penicillium* sp. was conducted. Our recent study reports the information on the capacity of *Penicillium* sp. Sj-2-2 producing GAs. This study emphasizes the importance of using *Penicillium* sp. for plant growth and development under salt stress environment. For the novel strain of *Penicillium* sp., further studies on field tests are recommended.

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